The implicit BTCS scheme and the Thomas algorithm

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1 Setting and motivation

The explicit FTCS scheme of the previous note is conditionally stable: $\alpha = (\sigma^2/2)\Delta\tau/\Delta x^2 \leq 1/2$ forces $M \gtrsim N^2$ time steps and $O(N^3)$ total work. This note implements BTCS (Backward-Time, Centred-Space), an *unconditionally stable* scheme: any $\Delta\tau > 0$ produces a convergent solution, decoupling M from N.

The cost of unconditional stability is that each time step requires solving a linear system. This note has therefore two deliverables:

1. the BTCS pricer for European options;
2. the *Thomas algorithm*, a specialised Gaussian elimination that solves tridiagonal systems in $O(N)$ work.

Thomas is implemented from scratch as a standalone module (`quantlib.thomas` in Python, `quant::pde::thomas` in C++). It is reused by the Crank–Nicolson pricer through the generic θ-scheme stepper (`quantlib.theta_scheme`); the American pricer replaces it with PSOR (`quantlib.psor`).

2 The BTCS discretisation

We discretise the transformed Black–Scholes PDE

$$\frac{\partial V}{\partial \tau} = \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V \tag{1}$$

on the grid of the finite-difference foundations note. BTCS evaluates every spatial stencil at the *new* time level $n+1$:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} = \frac{\sigma^2}{2}\,\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta x^2} + \mu\,\frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta x} - r\,V_j^{n+1}.$$

Multiplying by $\Delta\tau$ and grouping all V^{n+1} on the left, V^n on the right:

$$\boxed{b_-\,V_{j-1}^{n+1} + b_0\,V_j^{n+1} + b_+\,V_{j+1}^{n+1} \;=\; V_j^n} \tag{2}$$

with

$$b_- = -\alpha + \frac{\mu\,\Delta\tau}{2\,\Delta x}, \qquad b_0 = 1 + 2\alpha + r\,\Delta\tau, \qquad b_+ = -\alpha - \frac{\mu\,\Delta\tau}{2\,\Delta x},$$

where $\alpha = (\sigma^2/2)\Delta\tau/\Delta x^2$ as before.

Equation (2) is the FTCS scheme *written from the other side*: where FTCS reads "V_j^{n+1} equals an explicit combination of V^n", BTCS reads "V_j^n equals an explicit combination of V^{n+1}". The latter is a system to be solved for the unknowns V^{n+1}. The two schemes are the $\theta = 0$ and $\theta = 1$ extremes of the family

$$b_-\,V_{j-1}^{n+1} + b_0\,V_j^{n+1} + b_+\,V_{j+1}^{n+1} = a_-\,V_{j-1}^n + a_0\,V_j^n + a_+\,V_{j+1}^n,$$

the θ-scheme, with Crank–Nicolson at $\theta = 1/2$ (see the Crank–Nicolson note).

3 The tridiagonal system

For each time step, equation (2) written for the interior nodes $j = 1, \dots, N-1$ is a linear system $A\,\mathbf{V}_{\text{int}}^{n+1} = \mathbf{d}$ where $A \in \mathbb{R}^{(N-1)\times(N-1)}$ is tridiagonal:

$$A = \begin{pmatrix} b_0 & b_+ & & & \\ b_- & b_0 & b_+ & & \\ & b_- & b_0 & b_+ & \\ & & \ddots & \ddots & \ddots \\ & & & b_- & b_0 \end{pmatrix}.$$

The boundary nodes V_0^{n+1} and V_N^{n+1} are known from the Dirichlet conditions of the foundations note; they appear in the right-hand side:

$$\mathbf{d} = \begin{pmatrix} V_1^n - b_-\,V_0^{n+1} \\ V_2^n \\ \vdots \\ V_{N-2}^n \\ V_{N-1}^n - b_+\,V_N^{n+1} \end{pmatrix}. \tag{3}$$

The first and last rows pick up the contributions from the boundary; the rest of $\mathbf{d}$ is just V^n at the interior.

Constancy of A in time. The coefficients b_-, b_0, b_+ are independent of n and j thanks to the constant-coefficient form (1). The matrix A is therefore the same at every time step; only the right-hand side $\mathbf{d}$ changes. This is the structural pay-off of the logarithmic transformation that we already exploited in FTCS, now in a stronger form: at each step we solve a system with the *same* matrix and a different right-hand side. The matrix-dependent part of the forward sweep of the Thomas algorithm can be done *once* at setup and reused, cutting the per-step operation count from about $8n$ to about $5n$, a factor ≈ 1.6 (Section 5).

4 Stability: unconditional

The von Neumann analysis of (2) mirrors the foundations note, §9.3, extended with the convective and reactive terms. Substituting $V_j^n = g^n e^{ij\theta}$ and solving for g:

$$g(\theta) = \frac{1}{1 + 4\alpha \sin^2(\theta/2) - i\,\beta \sin\theta + r\,\Delta\tau}, \qquad \beta = \frac{\mu\,\Delta\tau}{\Delta x}. \tag{4}$$

The squared modulus,

$$|g|^2 = \frac{1}{\big(1 + 4\alpha \sin^2(\theta/2) + r\Delta\tau\big)^2 + \beta^2 \sin^2\theta},$$

has denominator ≥ 1 for every θ (every term is non-negative; one of them, 1, is strictly so). Hence $|g(\theta)| \leq 1$ unconditionally: BTCS is stable for any $\Delta\tau > 0$ and $\Delta x > 0$.

Combined with consistency ($O(\Delta\tau)+O(\Delta x^2)$ local truncation), the Lax equivalence theorem gives convergence:

$$\|V_h - V\|_\infty = O(\Delta\tau) + O(\Delta x^2).$$

Remark 4.1 (Stability and monotonicity)**.** *L^2 stability does not in general imply L^∞ monotonicity, but for BTCS both hold for every $\Delta\tau$ when $\mathrm{Pe}_h = |\mu|\Delta x/\sigma^2 \leq 1$. Since $\mu\Delta\tau/(2\Delta x) = \alpha\,\mu\Delta x/\sigma^2$, the off-diagonal entries are $b_\pm = -\alpha\,(1 \pm \mu\Delta x/\sigma^2)$: both α and the convective term*

scale linearly in $\Delta\tau$, so the signs of $b_\pm$ depend only on Pe_h, and both are non-positive when $\mathrm{Pe}_h \le 1$. With the strict diagonal dominance of Section 6 (margin $1 + r\Delta\tau$ for every $\Delta\tau$), A is then an M-matrix, $A^{-1} \ge 0$ entrywise, and the scheme is monotone: it creates no oscillations near the strike kink, however large $\Delta\tau$ is. Large-$\Delta\tau$ oscillations at the kink are a Crank–Nicolson phenomenon (see the Crank–Nicolson note). For BTCS a large $\Delta\tau$ costs only accuracy, through the $O(\Delta\tau)$ error: the large-$\Delta\tau$ tests in the validation give prices that are merely imprecise, never explosive.

5 The Thomas algorithm

Solving $A\mathbf{x} = \mathbf{d}$ with general Gaussian elimination costs $O(N^3)$ — prohibitive at $N = 200$, hopeless at $N = 1000$. The Thomas algorithm exploits the tridiagonal structure to compute the same solution in $O(N)$ work, which is asymptotically optimal: just reading the input takes $O(N)$ time.

Setup. Consider the general tridiagonal system

$$\begin{pmatrix} \tilde{b}_1 & c_1 & & & \\ a_2 & \tilde{b}_2 & c_2 & & \\ & a_3 & \tilde{b}_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & \tilde{b}_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix}.$$

We write $\tilde{b}$ for the diagonal of the general system to distinguish from the BTCS-specific b_0, b_-, b_+.

Forward sweep. Standard Gaussian elimination, applied naively to a tridiagonal system, produces *no fill*: at each step the only non-zero entries below the pivot are a_i, and the elimination touches $\tilde{b}_i$ and d_i but not the rows below. The recursion is

$$\begin{aligned} c'_1 &= \frac{c_1}{\tilde{b}_1}, & d'_1 &= \frac{d_1}{\tilde{b}_1}, \\ m_i &= \tilde{b}_i - a_i\, c'_{i-1}, & c'_i &= \frac{c_i}{m_i}, & d'_i &= \frac{d_i - a_i\, d'_{i-1}}{m_i}, \quad i = 2, \dots, n. \end{aligned} \tag{5}$$

The m_i are the *effective pivots*: what remains on the diagonal after eliminating a_i. After the sweep the system is upper bidiagonal:

$$\begin{pmatrix} 1 & c'_1 & & & \\ & 1 & c'_2 & & \\ & & 1 & c'_3 & \\ & & & \ddots & \end{pmatrix} \mathbf{x} = \mathbf{d}'.$$

Backward substitution. Trivial:

$$x_n = d'_n, \qquad x_i = d'_i - c'_i\, x_{i+1}, \quad i = n-1, n-2, \dots, 1. \tag{6}$$

Operation count. The forward sweep does roughly $6n$ floating-point operations (one division for c'_i, three operations for d'_i, two for m_i). The backward substitution does $\sim 2n$. Total: $O(n)$ with constant ≈ 8.

Memory. Only four vectors of length $\le n$: the three diagonals (modified in place if desired) and the right-hand side. No fill-in, no $O(n^2)$ storage.

Pre-factoring for repeated solves. The forward sweep (5) computes two distinct families of quantities: those depending only on $(a, \tilde{b}, c)$, namely $\{m_i, c'_i\}$, and those depending also on d, namely $\{d'_i\}$. When the matrix is constant across many right-hand sides — exactly our case, with one fixed A and M different right-hand sides — we compute the (m_i, c'_i) once at setup, and each subsequent solve costs only the d' recursion plus the backward substitution. The saving is the (m_i, c'_i) recursion, about $3n$ of the $8n$ operations per solve (including $n-1$ divisions), i.e. a factor $8/5 \approx 1.6$ in operation count.

6 Stability of Thomas without pivoting

The Thomas algorithm as written does *not* pivot: there is no provision for $m_i = 0$ or $|m_i|$ small. In general, Gaussian elimination without pivoting is unsafe — a small pivot amplifies round-off catastrophically. For Thomas there is a clean theorem that rules out the failure mode for our matrices.

Theorem 6.1 (Thomas without pivoting under diagonal dominance)**.** *If the tridiagonal matrix is strictly diagonally dominant by rows,*

$$|\tilde{b}_i| > |a_i| + |c_i|, \quad i = 1, \ldots, n \qquad (a_1 = c_n = 0), \tag{7}$$

then all effective pivots m_i in (5) satisfy $|m_i| \geq |\tilde{b}_i| - |a_i| > 0$, and the algorithm is backward stable: the computed solution $\hat{\mathbf{x}}$ satisfies $(A + \delta A)\hat{\mathbf{x}} = \mathbf{d}$ with $\|\delta A\|_\infty \leq c\,\varepsilon_{\text{mach}}\|A\|_\infty$ for a small constant c independent of n.

(Proof in Higham, *Accuracy and Stability of Numerical Algorithms*, §9.5–9.6.) Non-strict dominance with strict inequality in only one row is not enough: $\mathrm{diag}(1, 0)$ satisfies it and has a zero pivot. It does suffice for non-zero pivots if, in addition, the matrix is irreducible, i.e. all a_i $(i \geq 2)$ and c_i $(i \leq n-1)$ are non-zero.

Application to BTCS. The BTCS matrix has $\tilde{b}_i = b_0$, $a_i = b_-$, $c_i = b_+$, all constant in i. Diagonal dominance reduces to

$$|b_0| \;\geq\; |b_-| + |b_+|.$$

We have $|b_0| = 1 + 2\alpha + r\Delta\tau$ (positive). Computing the right-hand side:

$$|b_-| + |b_+| = \left|\alpha - \frac{\mu\Delta\tau}{2\Delta x}\right| + \left|\alpha + \frac{\mu\Delta\tau}{2\Delta x}\right|.$$

When $\alpha \geq \left|\mu\Delta\tau/(2\Delta x)\right|$ — i.e., $|\mu|\Delta x/\sigma^2 \leq 1$, the grid Péclet condition of the FTCS note — both terms inside the absolute values have α dominating and the same sign, and we get $|b_-| + |b_+| = 2\alpha$. Then

$$|b_0| - (|b_-| + |b_+|) = 1 + r\Delta\tau > 0,$$

strict diagonal dominance in every row (the first and last rows have only one off-diagonal entry). So Thomas without pivoting is provably stable for BTCS whenever $\mathrm{Pe}_h \leq 1$, which holds across the project's parameter range (the validation and benchmark grids have $\mathrm{Pe}_h \leq 0.025$; see the FTCS note).

7 Convergence rates and computational cost

The BTCS error is $O(\Delta\tau) + O(\Delta x^2)$. With M now independent of N, the user has a choice:

Quadratic-precision regime ($M \sim N^2$). Choose $\Delta\tau \sim \Delta x^2$. Both error contributions are $O(\Delta x^2)$; the total error is $O(\Delta x^2)$. The computational cost is $O(N) \cdot M \cdot 2 = O(N \cdot N^2) = O(N^3)$, with the constant a factor ~ 5 from the pre-factored Thomas count. *Same asymptotic cost as FTCS for the same precision.*

Linear-precision regime ($M \sim N$). Choose $\Delta\tau \sim \Delta x$. The error is dominated by $O(\Delta\tau) = O(\Delta x)$; the spatial $O(\Delta x^2)$ contribution is sub-dominant. The total error is $O(\Delta x)$. Cost: $O(N) \cdot M = O(N^2)$. *Acceptable when first-order precision is enough* (prototyping, calibration of insensitive parameters, sanity checks).

Crank–Nicolson preview. Crank–Nicolson achieves $O(\Delta\tau^2) + O(\Delta x^2)$ local truncation, so $M \sim N$ gives full $O(\Delta x^2)$ precision at $O(N^2)$ cost. That is the point where the explicit-vs-implicit trade decisively tips toward implicit.

Summary table.

Scheme	Local error	Stability	For $O(\Delta x^2)$	For $O(\Delta x)$
FTCS	$\Delta\tau + \Delta x^2$	$\alpha \leq 1/2$	$O(N^3)$ (forced)	not relevant
BTCS	$\Delta\tau + \Delta x^2$	unconditional	$O(N^3)$ ($M \sim N^2$)	$O(N^2)$ ($M \sim N$)
CN	$\Delta\tau^2 + \Delta x^2$	unconditional	$O(N^2)$ ($M \sim N$)	overkill

8 Implementation outline

The deliverables are two modules each in Python and C++:

- `quantlib.thomas` (Python) / `quant::pde::thomas_*` (C++) provides `thomas_solve(sub, diag, sup, rhs)` for one-shot solves and the pair `thomas_factor(sub, diag, sup)` + `thomas_solve_factored(factor, rhs)` for repeated solves with a fixed matrix (in C++, the second step is the method `ThomasFactor::solve(rhs)` of the returned factor).
- `quantlib.btcs` (Python) / `quant::pde::btcs_*` (C++) provides `btcs_european_call/put` with the same signature as the FTCS pricers, except that there is no CFL check: any $M \geq 1$ is valid input.

The validation suite checks five properties:

1. Cross-validation of `thomas_solve` against `numpy.linalg.solve` (Python) and against an independent dense LU implementation in the C++ tests, for randomly-generated diagonally dominant matrices.
2. Cross-validation of BTCS against Black–Scholes closed-form, with tolerance scaled to the discretisation error.
3. Convergence rate at $\Delta\tau \sim \Delta x^2$: error reduction by a factor of ~ 4 when $N \to 2N$, $M \to 4M$.
4. First-order convergence in time at fixed (large) N, $M \to 2M$: error reduction by ~ 2.
5. *Large-$\Delta\tau$ stability test*: BTCS at $M = 10$ for $T = 1$ produces a finite, sensible price (although imprecise), with no explosion. This is the operational signature of unconditional stability.

References

References

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